Problem B: Castle Invasion ($5 \times 3 = 15$ points)

By Joshua Lin

You have $N = 10$ castles, each with values $1, 2, \cdots, N$. Your task is to allocate $M = 100$ soldiers to castles. You play against an opponent. You win a castle if you allocate more soldiers to the castle than the opponent. You win a *match* if you win castles $\mathcal{N}$, your opponent wins castles $\mathcal{M}$, and $\sum_{n \in \mathcal{N}} n > \sum_{m \in \mathcal{M}} m$. Your total score for this problem is **proportional to your win-rate percentile**

In this game, every team will play against every other team. Teams can submit multiple times, but only the last submission prior to each deadline will be counted There will be three rounds, concluding at 1:00PM, 2:00PM, and 3:00PM respectively.

An Important Caveat!

However, after each round, this document will change, and the rules of Castle Invasion will be made slightly different. For example, the next round *could be* that, let $f(i)$ be the amount of troops you allocate to castle i, and let $g(i)$ represent your opponents allocation for castle i, you win a match if you win castles $\mathcal{N}$, your opponent wins castles $\mathcal{M}$, and $\sum_{n \in \mathcal{N}} n \cdot (f(i) - g(i))^2 > \sum_{m \in \mathcal{M}} m \cdot (g(i) - f(i))^2$.

Response Format

For each round, teams submit their allocation by Google Form. The forms are available HERE.

Note: after the end of each round, every team's submissions and rank ***will be made public****.*

Example

Team 1 submits an allocation,

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[0, 0, 0, 0, 0, 20, 20, 20, 20, 20]
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Team 2 submits an allocation,

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[10, 10, 10, 10, 10, 10, 10, 10, 10, 10]
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Team 1 wins castles $\mathcal{N} \equiv [6, 10]$. Team 2 wins castles $\mathcal{M} \equiv [1, 5]$. Hence

$$\sum_{n \in \mathcal{N}} n = 6 + 7 + 8 + 9 + 10 = 40, \quad \sum_{m \in \mathcal{M}} m = 1 + 2 + \cdots + 5 = 15$$

so since $40 > 15$, Team 1 wins the match.